

INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP 7.0

Workflow Report On

AI-Assisted Threat Detection Dashboard Using AWS Cloud and Machine Learning

Submitted in Partial Fulfillment of the Requirements

for the

Infosys Springboard Internship 7.0

Submitted By

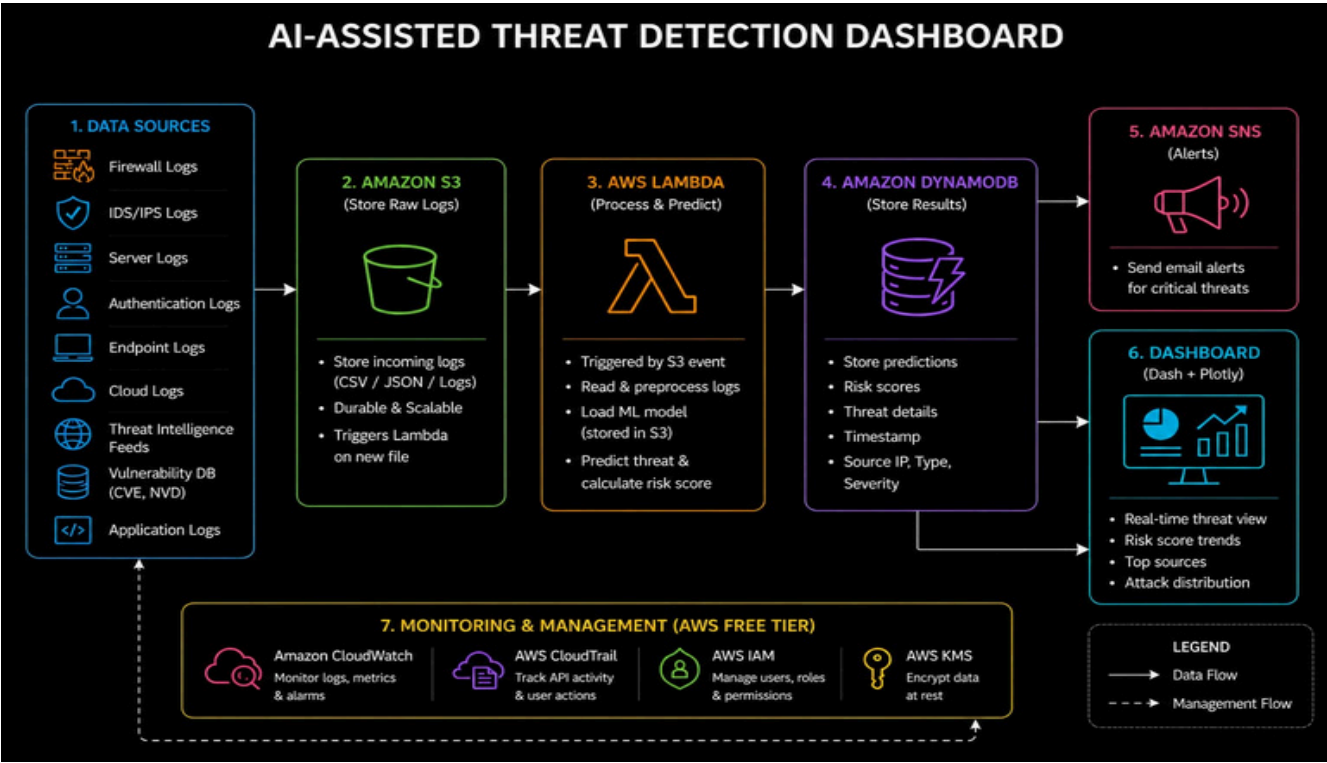
Name: Anjali Singh

Team: A (GROUP 2)

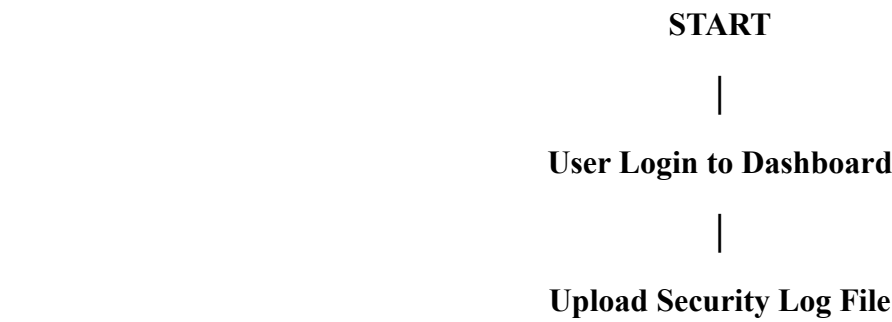
ARCHITECTURE DESCRIPTION

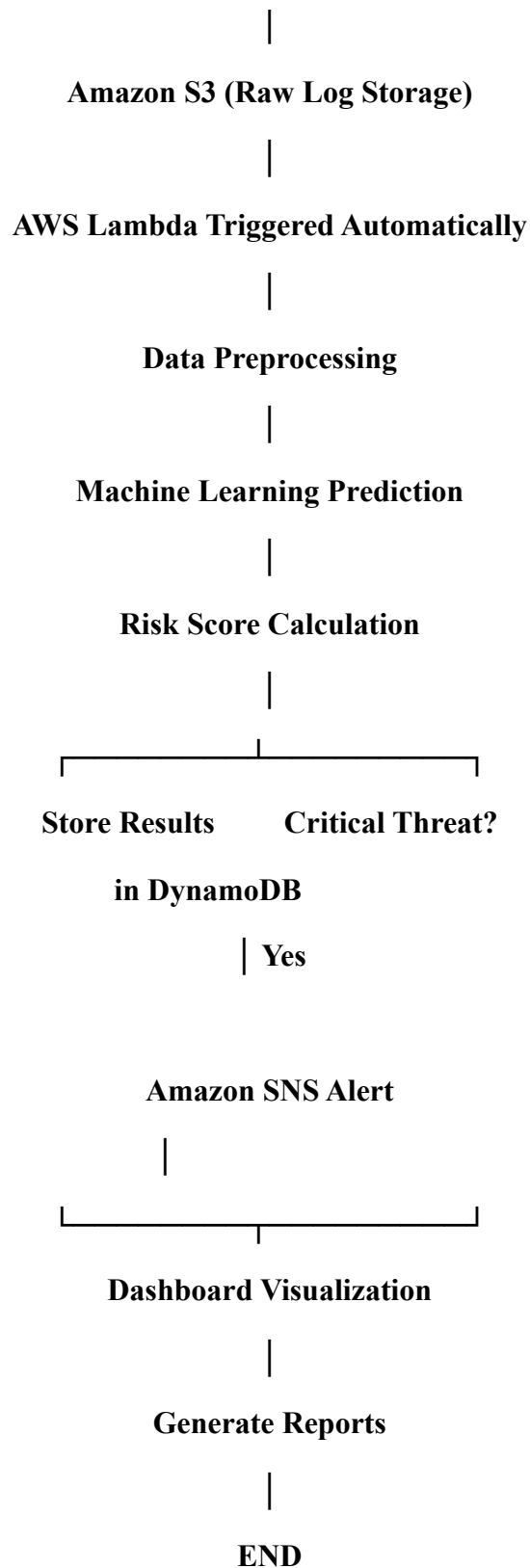
The proposed system follows a serverless AWS architecture to provide automated threat detection and visualization. Security logs are uploaded through the dashboard and stored in Amazon S3. An S3 event automatically invokes AWS Lambda, which preprocesses the log data and loads the trained Random Forest model to classify threats. The prediction results and calculated risk scores are stored in Amazon DynamoDB. If a critical threat is detected, Amazon SNS immediately sends an email notification to the administrator. Finally,

the dashboard retrieves data from DynamoDB and displays real-time security analytics, including threat summaries, risk scores, attack timelines, and downloadable reports.



Workflow Steps





Step 1: User Authentication

The user logs into the AI-Assisted Threat Detection Dashboard using valid credentials. After successful authentication, the user is redirected to the main dashboard.

Input:

- Username
- Password

Output:

- Dashboard Access

Step 2: Security Log Upload

The administrator uploads cybersecurity log files in CSV or JSON format through the dashboard. These logs may contain firewall events, authentication records, server logs, or network traffic data.

Input:

- CSV Log File
- JSON Log File

AWS Service Used:

- Amazon S3

Output:

- Raw logs stored securely in Amazon S3.

Step 3: Automatic Trigger

Whenever a new log file is uploaded to Amazon S3, AWS Lambda is automatically triggered using an S3 event notification.

AWS Service Used:

- AWS Lambda

Purpose:

- Start the threat detection process automatically.

Step 4: Data Preprocessing

AWS Lambda reads the uploaded log file and performs preprocessing before sending it to the Machine Learning model.

Operations Performed

- Remove missing values
- Validate log format
- Extract required features
- Convert categorical values
- Normalize data

Output:

- Cleaned security data ready for prediction.

Step 5: AI-Based Threat Detection

The preprocessed data is passed to the trained Machine Learning model developed using Scikit-learn.

Model Responsibilities

- Analyze login attempts
- Detect abnormal behavior
- Classify suspicious activities

- Predict cyber threats

Algorithm Used

- Random Forest Classifier

Output

- Normal Activity
- Malicious Activity

Step 6: Risk Score Generation

Based on the prediction, a risk level is assigned to every security event.

Risk Levels

Risk Score	Severity
0–25	Low
26–50	Medium
51–75	High
76–100	Critical

The risk score helps prioritize security incidents for investigation.

Step 7: Store Results

Prediction results, threat categories, and risk scores are stored in Amazon DynamoDB.

Stored Information

- Log ID
- Timestamp
- Source IP
- Threat Prediction
- Risk Score
- Threat Severity

AWS Service Used

- Amazon DynamoDB

Step 8: Alert Generation

If a threat is classified as **Critical**, AWS Lambda sends an alert notification to the administrator using Amazon SNS.

Alert Information

- Threat Type
- Risk Score
- Timestamp
- Recommended Action

AWS Service Used

- Amazon SNS

Step 9: Dashboard Visualization

The dashboard retrieves the latest prediction results from Amazon DynamoDB and displays them using interactive charts and tables.

Dashboard Components

- Total Threats
- Critical Alerts
- Risk Score
- Live Threat Table
- Threat Timeline
- Threat Distribution
- Recent Alerts
- Download Reports

Technology Used

- Dash (Python)
- Plotly